Top 30 Features vs fn_nano
(Within-Seq Spearman)

ρ=0.3
ρ=0.2

fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
glcm_contrast
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
top_edge_density
gradient_magnitude_std
laplacian_mean_abs
top_high_freq_energy
top_gradient_std
fft_critical_band_ratio
total_edge_components
small_edge_components
top_laplacian_var
glcm_entropy
laplacian_variance
mid_edge_density
fft_high_freq_energy_ratio
edge_fine_coarse_ratio
dark_channel_mean
mid_gradient_std
foreground_blob_count
glcm_energy
mid_high_freq_energy
rms_contrast

0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40

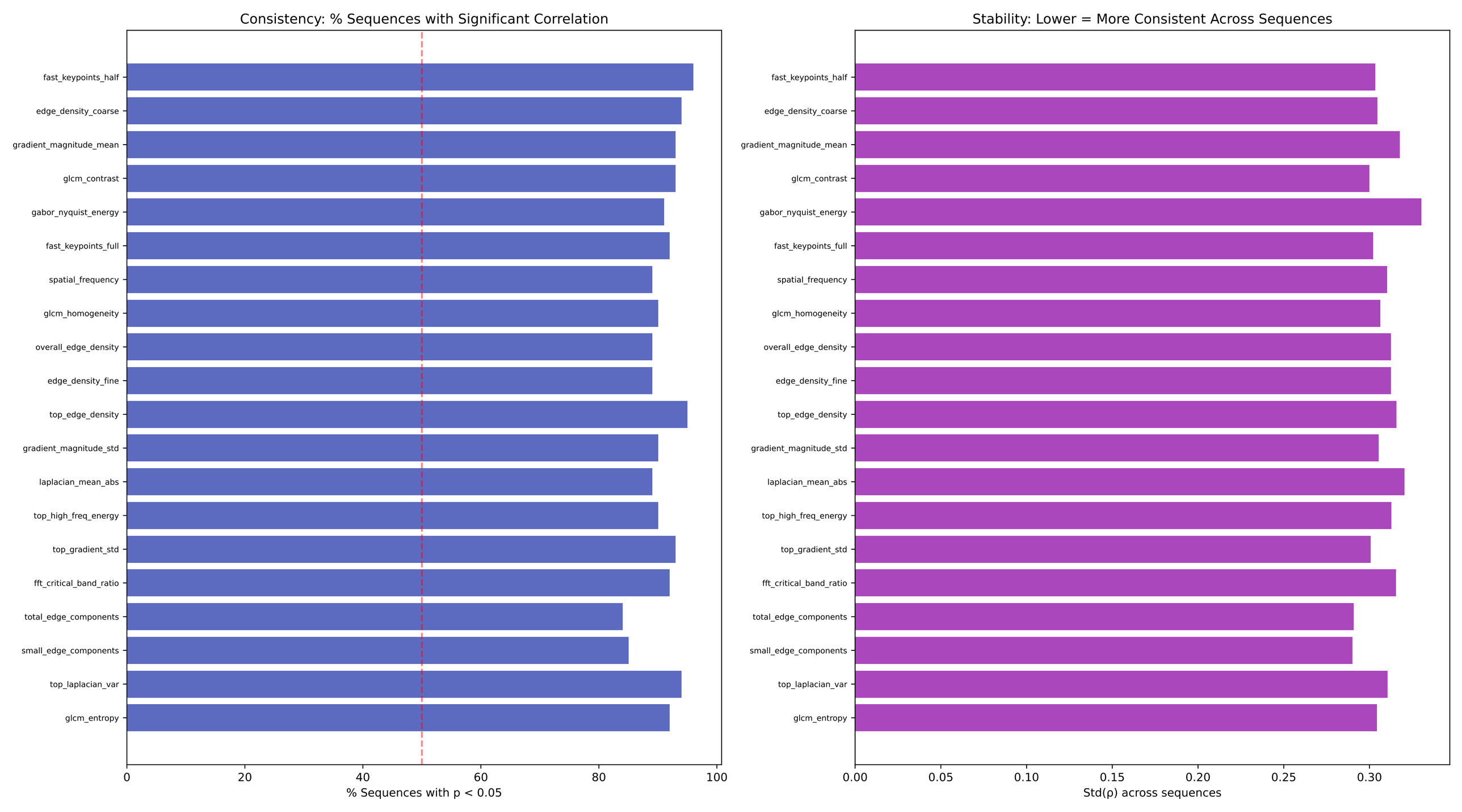
Mean |ρ| (within-sequence)

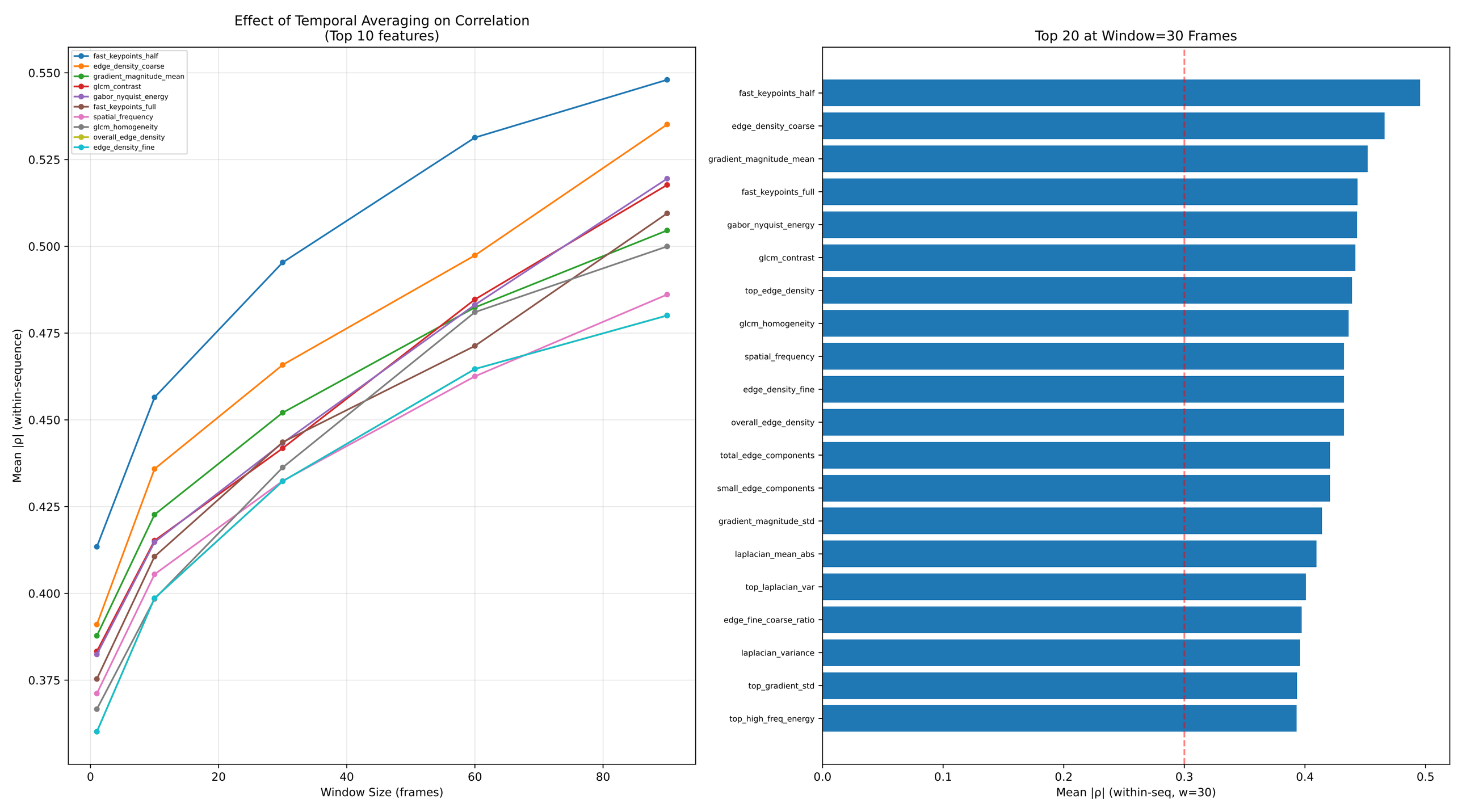
Direction of Correlation
(positive = more feature → more fn)

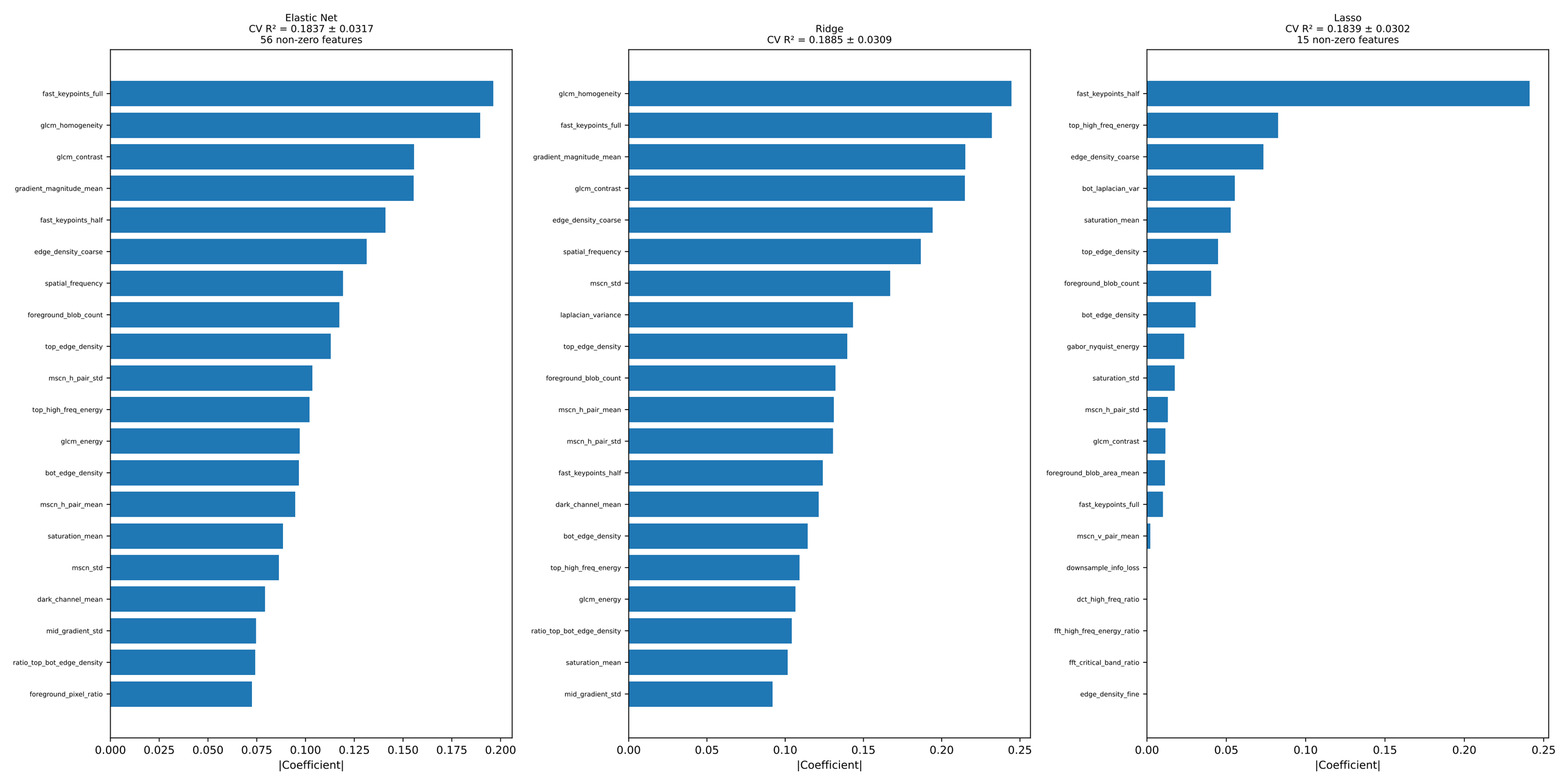
fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
glcm_contrast
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
top_edge_density
gradient_magnitude_std
laplacian_mean_abs
top_high_freq_energy
top_gradient_std
fft_critical_band_ratio
total_edge_components
small_edge_components
top_laplacian_var
glcm_entropy
laplacian_variance
mid_edge_density
fft_high_freq_energy_ratio
edge_fine_coarse_ratio
dark_channel_mean
mid_gradient_std
foreground_blob_count
glcm_energy
mid_high_freq_energy
rms_contrast

−0.3 −0.2 −0.1 0.0 0.1 0.2 0.3

Signed Mean ρ (within-sequence)


Consistency: % Sequences with Significant Correlation
fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
glcm_contrast
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
top_edge_density
gradient_magnitude_std
laplacian_mean_abs
top_high_freq_energy
top_gradient_std
fft_critical_band_ratio
total_edge_components
small_edge_components
top_laplacian_var
glcm_entropy
0
20
40
60
80
100
% Sequences with p < 0.05
Stability: Lower = More Consistent Across Sequences
fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
glcm_contrast
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
top_edge_density
gradient_magnitude_std
laplacian_mean_abs
top_high_freq_energy
top_gradient_std
fft_critical_band_ratio
total_edge_components
small_edge_components
top_laplacian_var
glcm_entropy
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Std(ρ) across sequences

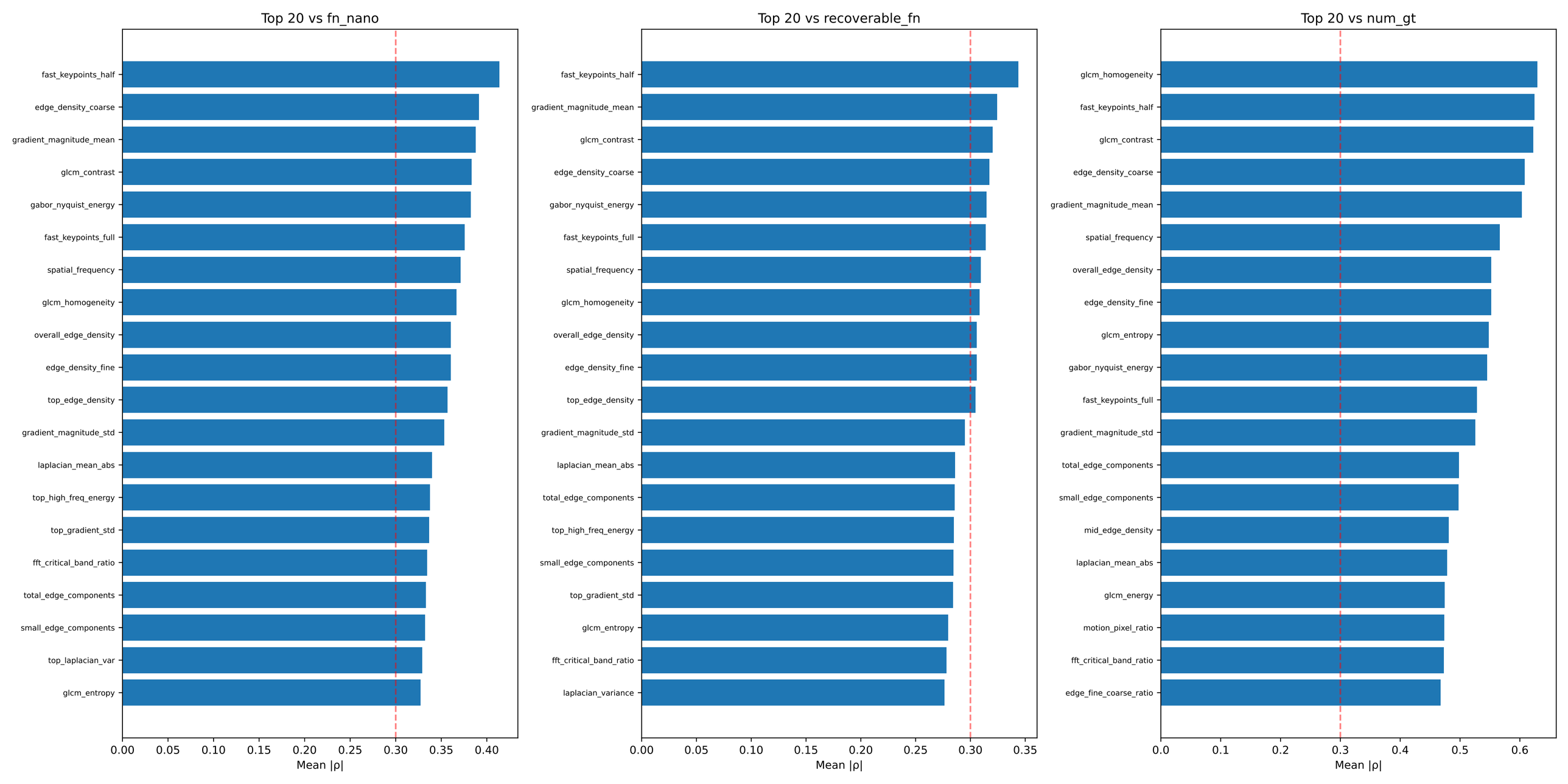

Effect of Temporal Averaging on Correlation
(Top 10 features)
fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
glcm_contrast
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
Mean |ρ| (within-sequence)
Window Size (frames)
0.550
0.525
0.500
0.475
0.450
0.425
0.400
0.375
0
20
40
60
80
Top 20 at Window=30 Frames
fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
fast_keypoints_full
gabor_nyquist_energy
glcm_contrast
top_edge_density
glcm_homogeneity
spatial_frequency
edge_density_fine
overall_edge_density
total_edge_components
small_edge_components
gradient_magnitude_std
laplacian_mean_abs
top_laplacian_var
edge_fine_coarse_ratio
laplacian_variance
top_gradient_std
top_high_freq_energy
0.0
0.1
0.2
0.3
0.4
0.5
Mean |ρ| (within-seq, w=30)


Elastic Net
CV R² = 0.1837 ± 0.0317
56 non-zero features
fast_keypoints_full
glcm_homogeneity
glcm_contrast
gradient_magnitude_mean
fast_keypoints_half
edge_density_coarse
spatial_frequency
foreground_blob_count
top_edge_density
mscn_h_pair_std
top_high_freq_energy
glcm_energy
bot_edge_density
mscn_h_pair_mean
saturation_mean
mscn_std
dark_channel_mean
mid_gradient_std
ratio_top_bot_edge_density
foreground_pixel_ratio
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
|Coefficient|
Ridge
CV R² = 0.1885 ± 0.0309
glcm_homogeneity
fast_keypoints_full
gradient_magnitude_mean
glcm_contrast
edge_density_coarse
spatial_frequency
mscn_std
laplacian_variance
top_edge_density
foreground_blob_count
mscn_h_pair_mean
mscn_h_pair_std
fast_keypoints_half
dark_channel_mean
bot_edge_density
top_high_freq_energy
glcm_energy
ratio_top_bot_edge_density
saturation_mean
mid_gradient_std
0.00
0.05
0.10
0.15
0.20
0.25
|Coefficient|
Lasso
CV R² = 0.1839 ± 0.0302
15 non-zero features
fast_keypoints_half
top_high_freq_energy
edge_density_coarse
bot_laplacian_var
saturation_mean
top_edge_density
foreground_blob_count
bot_edge_density
gabor_nyquist_energy
saturation_std
mscn_h_pair_std
glcm_contrast
foreground_blob_area_mean
fast_keypoints_full
mscn_v_pair_mean
downsample_info_loss
dct_high_freq_ratio
fft_high_freq_energy_ratio
fft_critical_band_ratio
edge_density_fine
0.00
0.05
0.10
0.15
0.20
0.25
|Coefficient|

Within-Sequence Feature Correlation (Mean |ρ|)
Top 25 Features + Target

fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
glcm_contrast
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
top_edge_density
gradient_magnitude_std
laplacian_mean_abs
top_high_freq_energy
top_gradient_std
fft_critical_band_ratio
total_edge_components
small_edge_components
top_laplacian_var
glcm_entropy
laplacian_variance
mid_edge_density
fft_high_freq_energy_ratio
edge_fine_coarse_ratio
dark_channel_mean
fn_nano

1.0
0.8
0.6
0.4
0.2
0.0


Top 20 vs fn_nano
fast_keypoints_half
edge_density_coarse
gradient_magnitude_mean
glcm_contrast
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
top_edge_density
gradient_magnitude_std
laplacian_mean_abs
top_high_freq_energy
top_gradient_std
fft_critical_band_ratio
total_edge_components
small_edge_components
top_laplacian_var
glcm_entropy
0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40
Mean |ρ|
Top 20 vs recoverable_fn
fast_keypoints_half
gradient_magnitude_mean
glcm_contrast
edge_density_coarse
gabor_nyquist_energy
fast_keypoints_full
spatial_frequency
glcm_homogeneity
overall_edge_density
edge_density_fine
top_edge_density
gradient_magnitude_std
laplacian_mean_abs
total_edge_components
top_high_freq_energy
small_edge_components
top_gradient_std
glcm_entropy
fft_critical_band_ratio
laplacian_variance
0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35
Mean |ρ|
Top 20 vs num_gt
glcm_homogeneity
fast_keypoints_half
glcm_contrast
edge_density_coarse
gradient_magnitude_mean
spatial_frequency
overall_edge_density
edge_density_fine
glcm_entropy
gabor_nyquist_energy
fast_keypoints_full
gradient_magnitude_std
total_edge_components
small_edge_components
mid_edge_density
laplacian_mean_abs
glcm_energy
motion_pixel_ratio
fft_critical_band_ratio
edge_fine_coarse_ratio
0.0 0.1 0.2 0.3 0.4 0.5 0.6
Mean |ρ|

```
================================================================================
COMPREHENSIVE FEATURE ANALYSIS SUMMARY
================================================================================

Dataset: DETRAC — 100 sequences, 138,252 frames
Features: 65
Primary target: fn_nano
Analysis: class-agnostic, within-sequence

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SPEARMAN TOP 5 (per-frame, mean |ρ|):
  fast_keypoints_half                    |ρ|=0.4134  signed=+0.3619  sig=96%
  edge_density_coarse                    |ρ|=0.3910  signed=+0.3353  sig=94%
  gradient_magnitude_mean                |ρ|=0.3878  signed=+0.3212  sig=93%
  glcm_contrast                          |ρ|=0.3833  signed=+0.3282  sig=93%
  gabor_nyquist_energy                   |ρ|=0.3824  signed=+0.3041  sig=91%

SPEARMAN TOP 5 (w=30 avg, mean |ρ|):
  fast_keypoints_half                    |ρ|=0.4954
  edge_density_coarse                    |ρ|=0.4658
  gradient_magnitude_mean                |ρ|=0.4521
  fast_keypoints_full                    |ρ|=0.4436
  gabor_nyquist_energy                   |ρ|=0.4433

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LINEAR MODEL COMPARISON (within-seq normalized, GroupKFold CV):
  elastic_net      R²=0.1837 ± 0.0317  (56 features)
  ridge            R²=0.1885 ± 0.0309
  lasso            R²=0.1839 ± 0.0302  (15 features)

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FEATURE CATEGORY PERFORMANCE:
  A: Resolution         n=17  best |ρ|=0.4134  mean |ρ|=0.3346
  B: Density            n=10  best |ρ|=0.2941  mean |ρ|=0.2233
  C: Complexity         n=15  best |ρ|=0.3833  mean |ρ|=0.3071
  D: Strip              n=15  best |ρ|=0.3569  mean |ρ|=0.2753
  E: BRISQUE            n= 8  best |ρ|=0.2727  mean |ρ|=0.2532

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REDUNDANCY CHECK (feature pairs with |ρ| > 0.9):
  edge_density_fine                  ↔ gradient_magnitude_mean             |ρ|=0.901
  edge_density_fine                  ↔ overall_edge_density                |ρ|=1.000
  glcm_contrast                      ↔ spatial_frequency                   |ρ|=0.900
  glcm_energy                        ↔ glcm_entropy                        |ρ|=0.931
  gradient_magnitude_mean            ↔ overall_edge_density                |ρ|=0.901
  gradient_magnitude_mean            ↔ spatial_frequency                   |ρ|=0.943
  gradient_magnitude_std             ↔ spatial_frequency                   |ρ|=0.978
  laplacian_mean_abs                 ↔ laplacian_variance                  |ρ|=0.944
  small_edge_components              ↔ total_edge_components               |ρ|=0.999
```